



Secret Management & GitOps Security

Enterprise-grade secret management, drift detection, self-healing, and progressive delivery for **Amazon EKS Anywhere on VMware** — managed by ArgoCD.

ARGOCD · EKS ANYWHERE · VMWARE



Secret Management Challenges

Never store secrets inside Git. Committing sensitive data to a repository exposes your entire organization to catastrophic risk — even in private repos.

Passwords & Keys

Hardcoded credentials leak across forks, clones, and CI logs.

Certificates & Tokens

Short-lived tokens and TLS certs require rotation — Git history never forgets.

⚖ Compliance & Risk

SOC 2, PCI-DSS, and HIPAA mandates prohibit plaintext secrets in version control.



Secret Management Options Compared

Choosing the right secret management tool for your EKS Anywhere on VMware environment depends on security posture, operational maturity, and enterprise readiness.

Solution	Security	Encryption	Ease of Use	Enterprise
Kubernetes Secrets	Basic	Base64 only		
Bitnami Sealed Secrets	Good	Asymmetric		Partial
Mozilla SOPS	Good	KMS/PGP		Partial
External Secrets Op.	Strong	Provider-native		
AWS Secrets Manager	Strong	AWS KMS		
HashiCorp Vault	Enterprise	Transit Engine		

ArgoCD Secret Integration on EKS Anywhere

Integration Patterns

01

Sealed Secrets

Encrypt in Git; controller decrypts on-cluster.

02

Vault Plugin

ArgoCD fetches secrets from Vault at sync time.

03

External Secrets

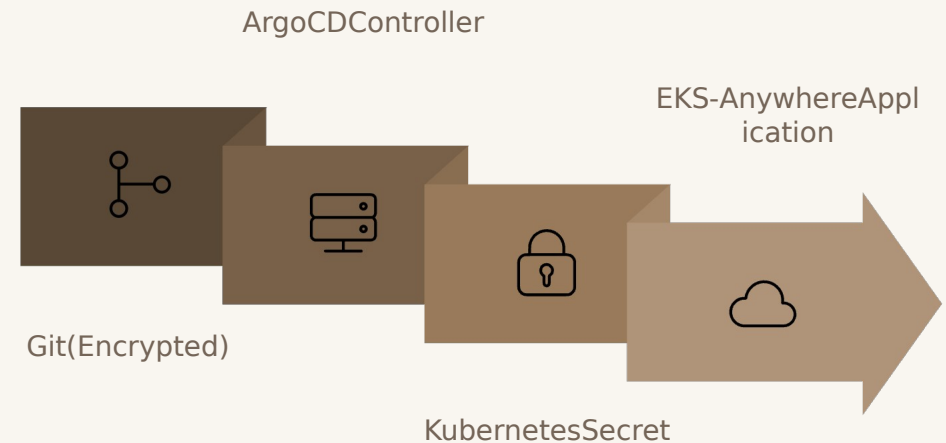
Operator syncs AWS Secrets Manager → K8s Secret.

04

Secret Rotation

Automated rotation without redeployment.

Architecture Flow



Configuration Drift: What It Is

Drift occurs when the *actual state* of your EKS Anywhere cluster diverges from the *desired state* declared in Git — silently undermining reliability and compliance.

Manual kubectl Edits

Ad-hoc changes applied directly to the cluster bypass Git entirely.

Emergency Fixes

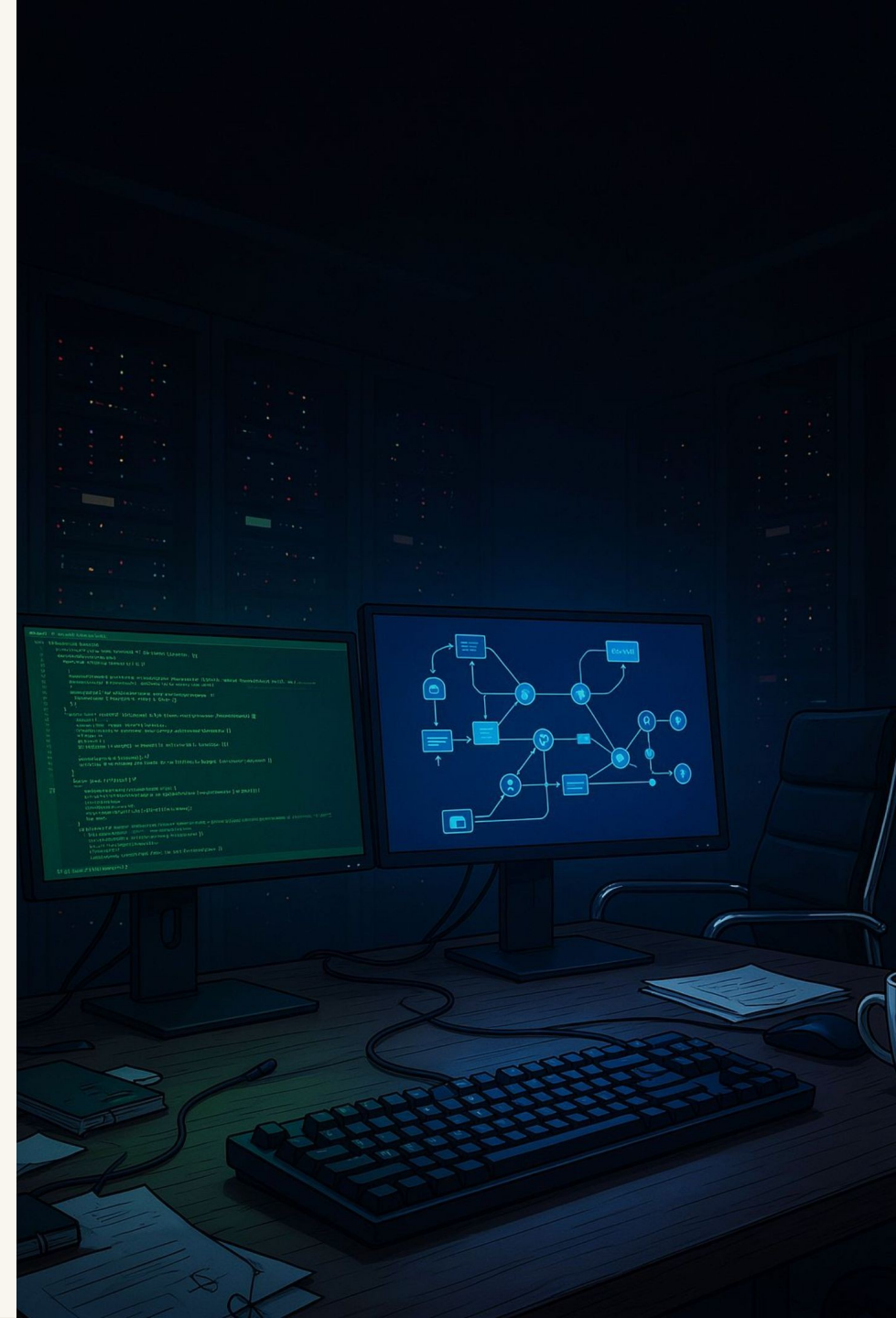
Hotfixes applied under pressure are rarely committed back to source.

Operator Changes

VMware or infrastructure operators modifying resources outside GitOps flow.

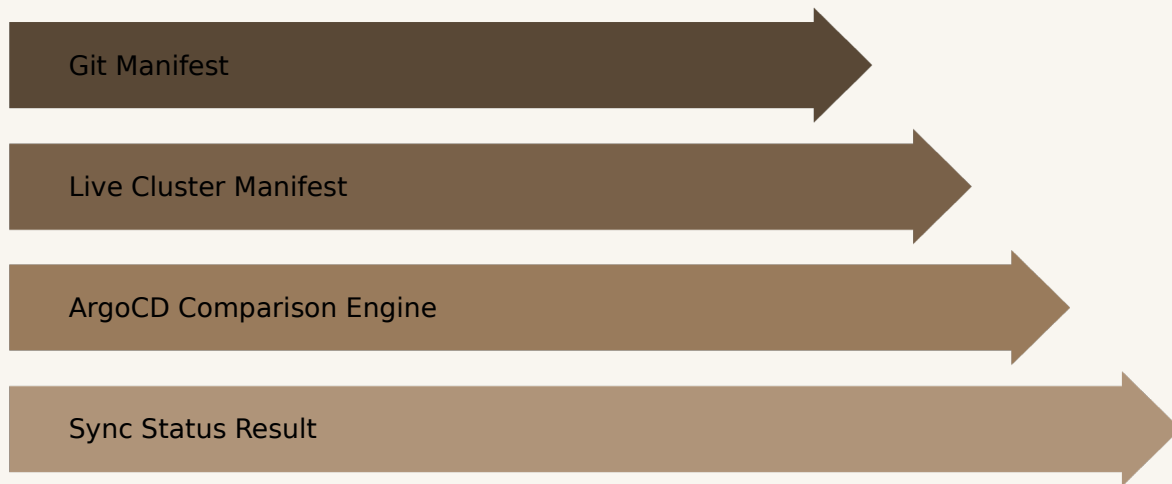
Infrastructure Updates

Node upgrades or platform patches that alter running workload configs.



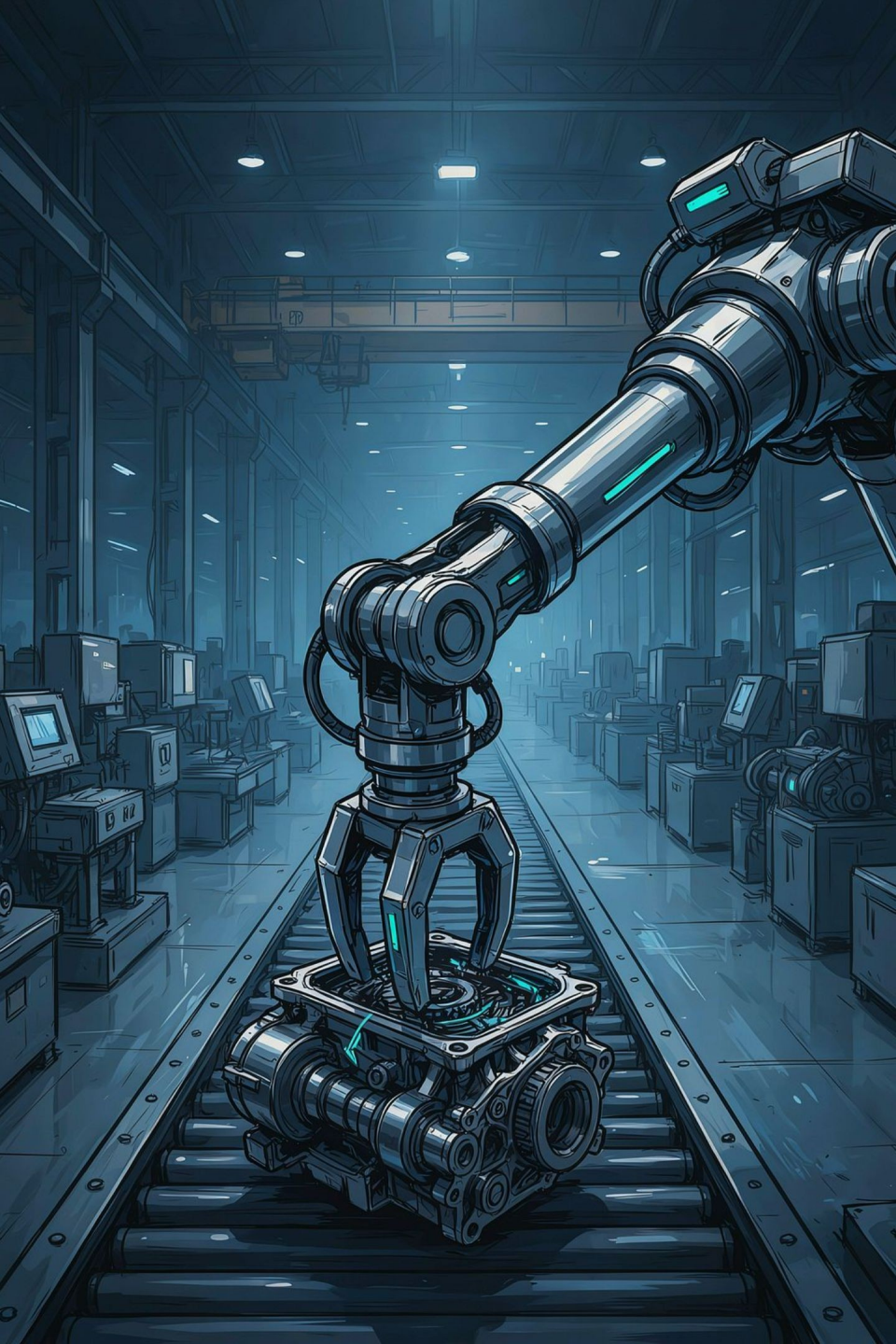
How ArgoCD Detects Drift

ArgoCD continuously compares the **live manifest** running in your VMware-hosted EKS Anywhere cluster against the **Git manifest** — flagging any divergence immediately.



Sync Status States

Healthy Cluster matches Git exactly.	⚠ OutOfSync Drift detected — reconciliation needed.
Unknown Health check cannot be determined.	Missing / Degraded Resource absent or failing health checks.



Self-Healing with ArgoCD

ArgoCD's self-healing capability automatically reconciles drift in EKS Anywhere clusters — restoring the desired Git state without human intervention.

Auto Heal

Detects OutOfSync state and triggers automatic sync back to Git truth.

Prune & Replace

Removes orphaned resources and replaces corrupted manifests cleanly.

Retry Logic

Configurable backoff retries handle transient VMware infrastructure failures.



Compliance

Enforces policy-compliant state continuously.



Security

Prevents unauthorized configuration changes from persisting.

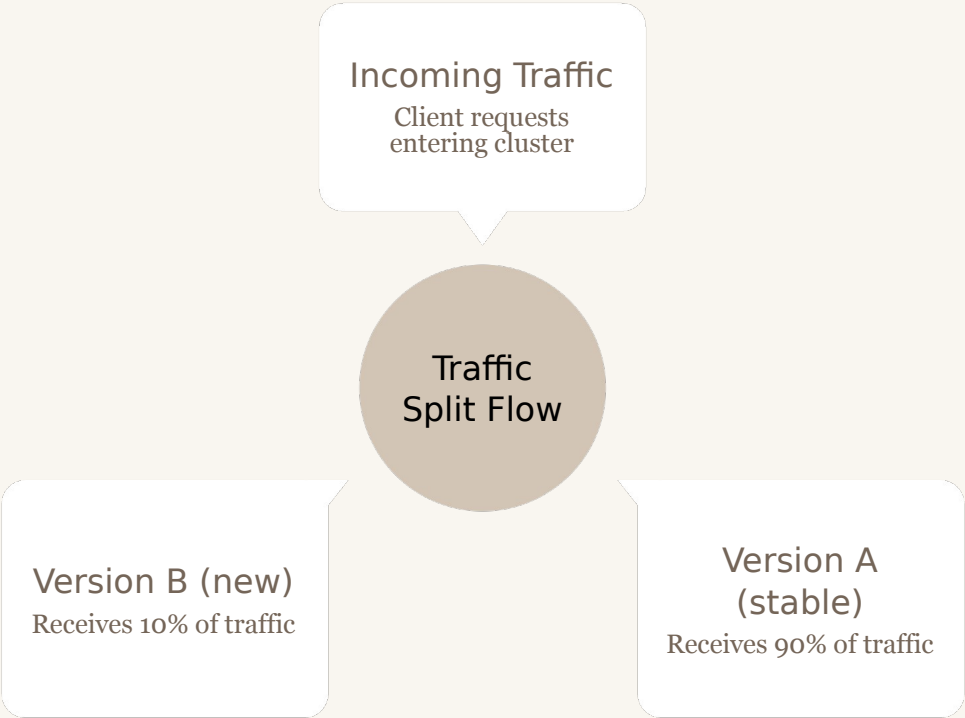
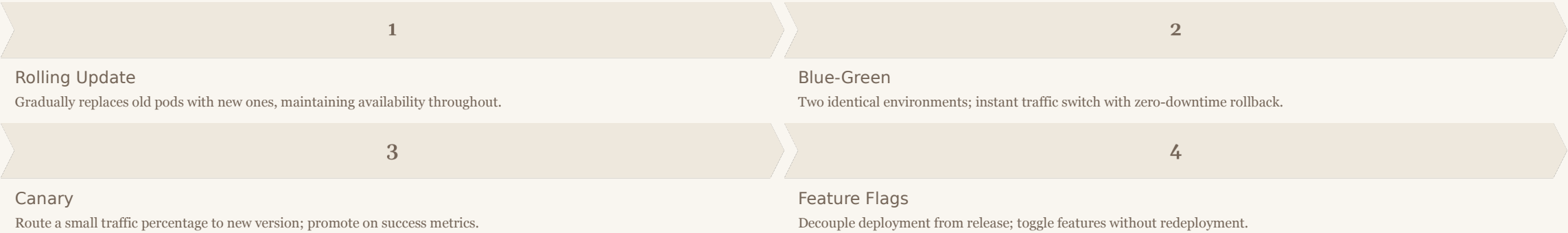


Disaster Recovery

Rapid cluster restoration from Git source of truth.

Progressive Delivery Strategies

Progressive delivery moves beyond traditional all-at-once deployments, enabling controlled, risk-managed rollouts on EKS Anywhere VMware clusters.



Canary Deployments in Detail

Traffic Progression

1

10% → Canary

Initial validation with minimal blast radius.

2

25% → Expand

Metrics healthy — widen the canary cohort.

3

50% → Majority

Half of production traffic on new version.

4

100% → Promote

Full promotion; old version decommissioned.

Key Metrics & Rollback

Error Rate

Threshold breach triggers automatic rollback.

Latency (p99)

Latency spikes halt promotion immediately.

CPU / Memory

Resource saturation signals instability.

Business KPIs

Conversion rate and revenue metrics gate promotion.

Blue-Green Deployment

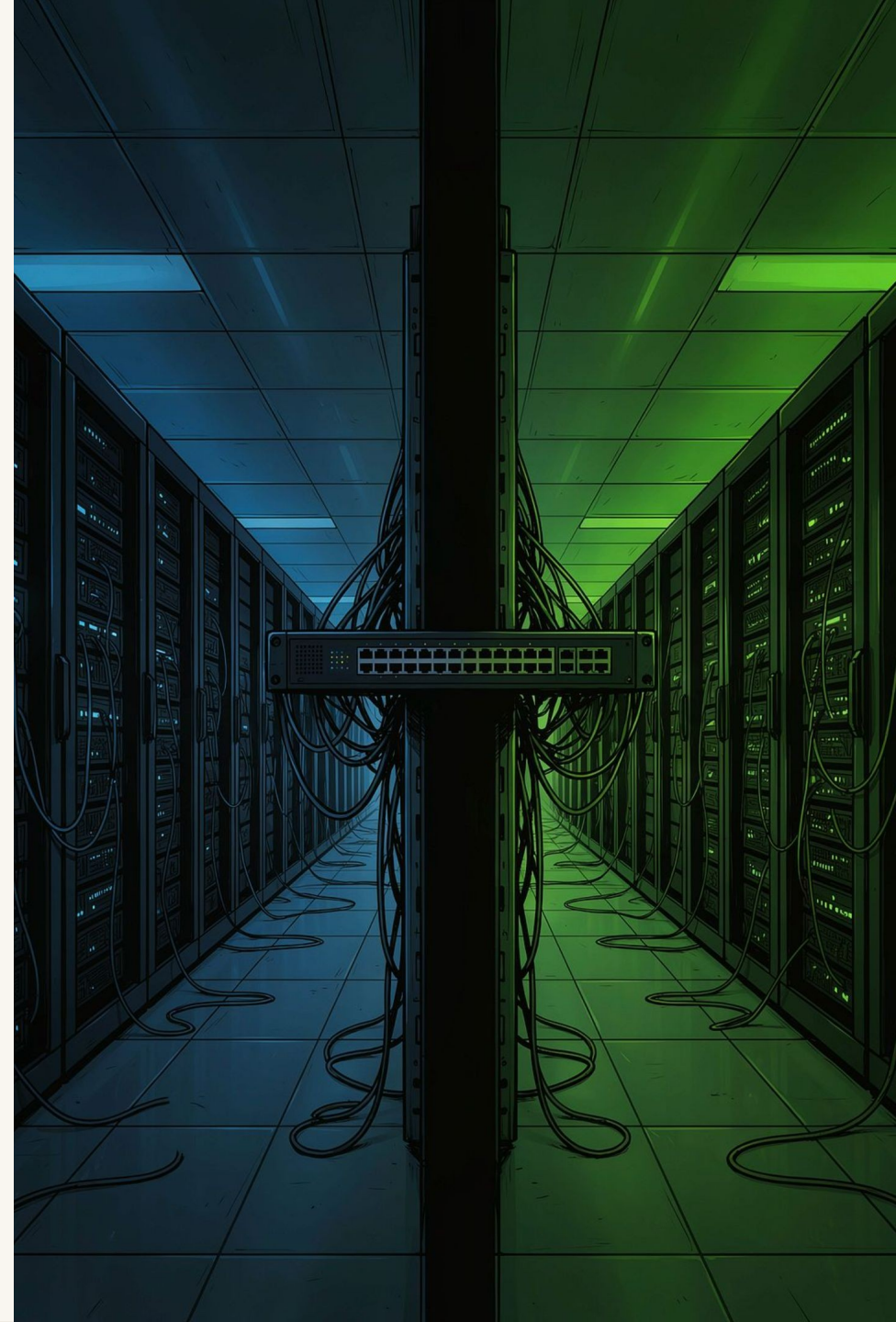
Blue-Green maintains two identical environments on EKS Anywhere VMware clusters. The load balancer switches traffic instantly — enabling zero-downtime releases and one-command rollback.

Advantages

- Instant, zero-downtime traffic switch
- Full rollback in seconds — just re-point the LB
- Green environment fully tested before promotion
- Eliminates partial-state deployment risk

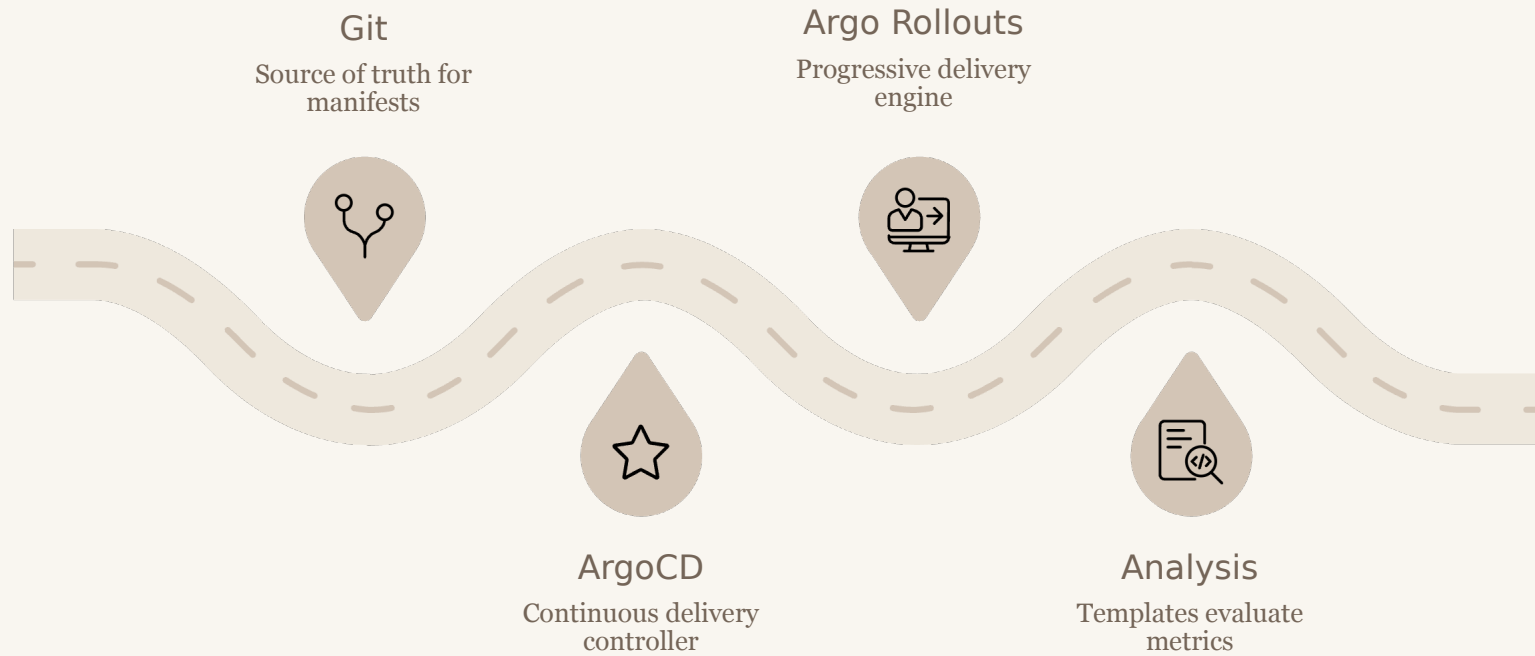
⚠ Disadvantages

- Requires 2× infrastructure capacity on VMware
- Database schema migrations need careful coordination
- Higher cost during transition window
- Session state must be externalized



Progressive Delivery with Argo Rollouts

Argo Rollouts extends ArgoCD with advanced delivery primitives — enabling automated canary and blue-green promotions driven by real-time metrics on EKS Anywhere.



Analysis Templates

Define metric thresholds that gate each promotion step automatically.

Experiments

Run A/B traffic experiments with statistical significance checks.

Pause & Promote

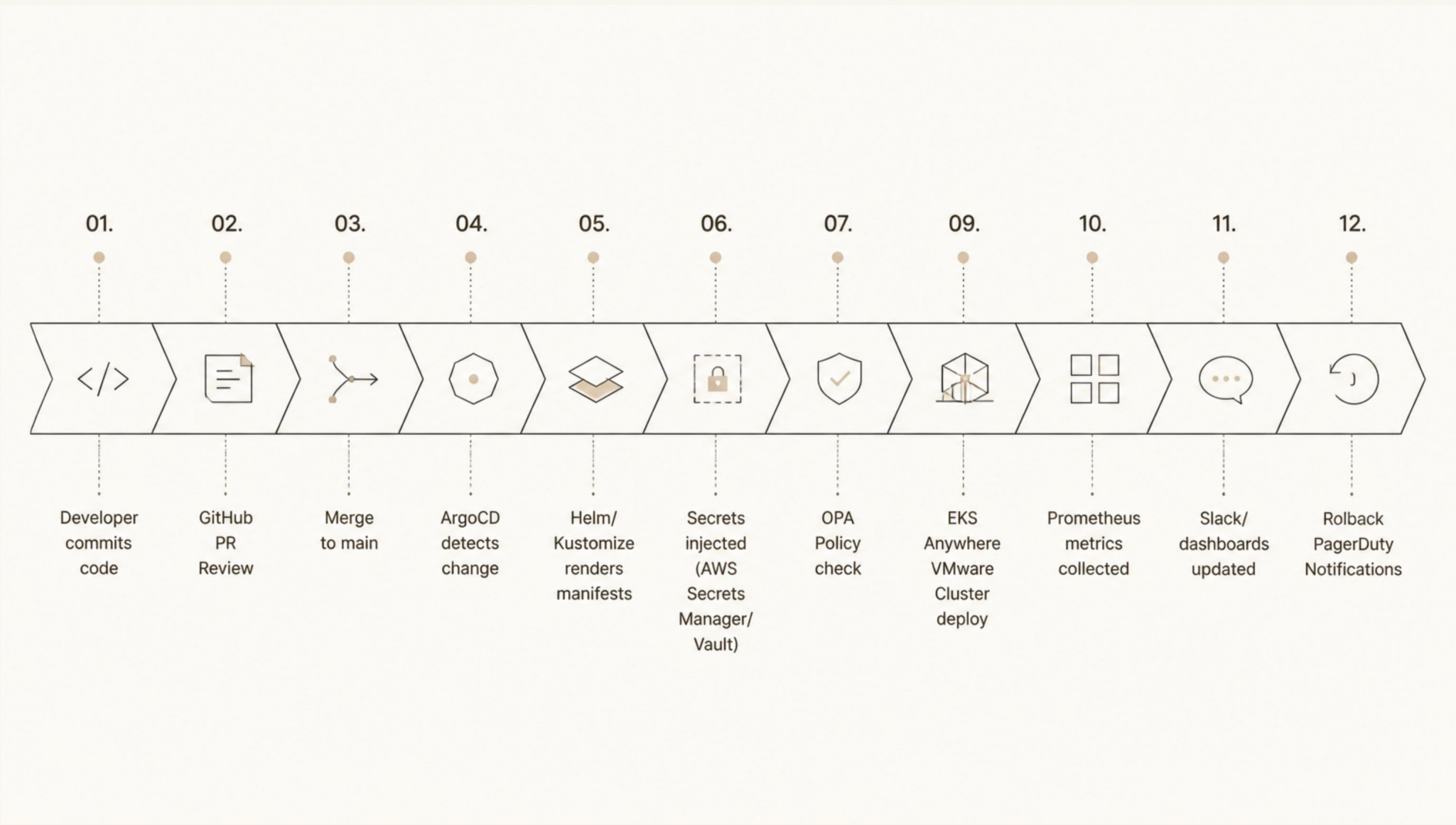
Manual approval gates or fully automated promotion on metric success.

Instant Rollback

Automatic rollback on analysis failure — no manual intervention required.

Complete Enterprise GitOps Pipeline

End-to-end GitOps workflow for EKS Anywhere on VMware — from developer commit to production, with full observability and automated rollback.



Production Best Practices: Repository & Access

Repository Design

1 Mono-repo vs. Multi-repo

Separate app code from infra manifests; use App-of-Apps pattern for EKS Anywhere clusters.

2 RBAC & App Projects

Scope ArgoCD Projects per team; restrict source repos and destination namespaces.

3 Sync Windows

Define maintenance windows to prevent deployments during peak VMware workload hours.

Health & Resource Controls

1 Health Checks & Resource Limits

Custom health checks per CRD; enforce CPU/memory limits on all workloads.

2 Namespace Isolation

Network policies and namespace boundaries prevent lateral movement on VMware nodes.

3 Policy Enforcement

OPA/Gatekeeper policies block non-compliant manifests before cluster admission.

Production Best Practices: Security & Audit



Signed Commits

Enforce GPG-signed commits on all GitOps repos. Only verified authors can trigger ArgoCD syncs to EKS Anywhere clusters.



Image Signing

Use Cosign/Notary to sign container images. ArgoCD verifies signatures before deploying to VMware-hosted nodes.



Secret Rotation

Automate rotation via AWS Secrets Manager + External Secrets Operator. Zero-downtime rotation without pod restarts.



Audit Logs

ArgoCD audit trail + Kubernetes audit logs shipped to SIEM. Full traceability for SOC 2 and PCI-DSS compliance.



Key Takeaways

Building a production-grade GitOps platform on **Amazon EKS Anywhere** with **VMware** requires disciplined secret management, continuous drift detection, and progressive delivery — all orchestrated by ArgoCD.

Never Commit Secrets

Use Sealed Secrets, Vault, or AWS Secrets Manager with External Secrets Operator.

Detect & Heal Drift

ArgoCD self-healing enforces Git as the single source of truth on every VMware node.

Ship Progressively

Argo Rollouts canary and blue-green strategies minimize blast radius on production clusters.

Enforce at Scale

RBAC, policy gates, signed commits, and audit logs are non-negotiable for enterprise EKS Anywhere.

